Heterogeneous Households and Aggregate Dynamics: Belief-Invariant Analytical Moment Recursion[†]

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Abstract

In a heterogeneous-household economy with idiosyncratic productivity and return risk, aggregate capital and two wealth-sorting covariances are sufficient for all contemporaneous allocations. Aggregate capital follows an exact closed-form law of motion, household policies are available in closed form, and equilibrium is belief-invariant under log preferences. The result gives an analytical counterpart to approximate aggregation in Krusell and Smith (1998): mean wealth is nearly sufficient when sorting is small, and exactly sufficient when it vanishes. An application to U.S. data validates the inversion on simulated Krusell–Smith economies, uncovers a two-dimensional business cycle, and uses belief invariance for rolling forecasts that outperform a recursive AR benchmark across major aggregates.

Keywords: Heterogeneous agents, closed-form aggregation, wealth inequality, business cycles, belief invariance.

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1 Introduction

Heterogeneous-agent macroeconomics is built around a tension. Distributional states are high dimensional, but aggregate dynamics often appear to depend on a small number of statistics. The most influential example is Krusell and Smith (1998): in a rich incomplete-markets model, mean capital nearly forecasts aggregate prices. That finding is computationally powerful but leaves two analytical questions open. Which objects does the representative-agent state omit? And when is omitting them innocuous?

This paper answers those questions in a tractable heterogeneous-household economy with idiosyncratic productivity and return risk. I show that the distribution enters contemporaneous equilibrium only through aggregate capital and two covariances: the alignment of wealth with labor productivity and the alignment of wealth with return heterogeneity. These covariances are not generic moments. They are the exact cross-sectional averages of the basis functions in the household saving rule. They measure sorting, and they are the objects through which inequality feeds back into aggregates.

The first contribution is an analytical moment-recursive competitive equilibrium (MRCE). Household consumption, saving, and labor supply are available in closed form. Given the moment vector

$$\mathcal{M} = \big[K,\ \mathrm{cov}(z,a),\ \mathrm{cov}(e,a)\big],$$

all contemporaneous prices and allocations are pinned down, and aggregate capital follows an exact law of motion. The result gives an analytical counterpart to approximate aggregation. Capital alone is exactly sufficient only when the sorting covariances vanish; it is approximately sufficient when those covariances are small relative to aggregate wealth. Shutting down return heterogeneity yields the Krusell–Smith analogue; shutting down all heterogeneity recovers the log growth model of Brock and Mirman (1972); with return risk alone, the distribution drops out as in Angeletos (2007).

The second contribution is belief invariance. Under log preferences, the income and substitution effects of future aggregate returns cancel, so the household policy functions depend on current prices but not on beliefs about the future law of motion for aggregates. The MRCE is therefore also a belief-invariant recursive competitive equilibrium. This is a

narrow but useful scope result: for this class of economies, one can compute non-stationary equilibrium paths by forward simulation using exact household policies, without solving a fixed point over perceived aggregate forecasting rules.

The third contribution is to separate what is exact from what is not. Aggregate capital is exactly recursive in the three moments, and the three moments are sufficient for contemporaneous equilibrium. The future covariances, however, generally depend on higher-order cross-sectional moments. The full distribution must therefore be tracked forward in quantitative work. This limitation is economically informative: the analytical result identifies the sufficient moment state for today's allocation, while the agent simulation records how the underlying distribution produces tomorrow's sorting.

I use the structure for a disciplined application to U.S. annual data from 1950 to 2019. The inversion recovers TFP, a saving-margin wedge, and an aggregate labor-efficiency wedge from output, capital, and hours. Because three shocks rationalize three observables, exact fit is not the test. The discipline comes from non-targeted implications. Applied to simulated Krusell–Smith economies, the inversion recovers TFP and leaves the other two shocks nearly constant. In U.S. data, the recovered labor-efficiency wedge is strongly correlated with external labor-market margins, the saving-margin wedge aligns with outside saving-side indicators, and the recovered shocks reveal a two-dimensional business cycle: TFP is nearly orthogonal to a labor-saving block. Belief invariance also makes conditional forecasting direct; a rolling exercise beats a recursive AR benchmark across major aggregates. The distributional simulation produces wealth concentration in the range observed in the Survey of Consumer Finances and illustrates the central mechanism: TFP growth raises returns, and persistent return-wealth sorting compounds those gains at the top. At the baseline calibration the resulting state-dependent amplification is modest, which is a useful scope condition rather than a headline magnitude.

Related literature. Bewley (1986), Huggett (1993), and Aiyagari (1994) developed the foundational heterogeneous-agent framework. Krusell and Smith (1998) showed numerically that mean wealth nearly suffices for aggregate forecasting; the MRCE identifies the omitted moments and gives exact conditions under which the approximation succeeds. The

result is closely related to Den Haan and Rendahl (2010), who show that cross-sectional averages of individual-policy basis functions are natural aggregate states. Here those averages reduce to economically interpretable sorting covariances. Computational approaches to high-dimensional heterogeneous-agent dynamics include Reiter (2009), Winberry (2018), Ahn et al. (2018), and the sequence-space Jacobian method of Auclert et al. (2021). The contribution here is complementary: it gives exact nonlinear dynamics for a restricted class rather than linearized dynamics for a broad class.

The paper also builds on exact aggregation under log preferences. Brock and Mirman (1972) solve the representative-agent growth model, and Angeletos (2007) obtains aggregation with uninsured return risk. This paper shows how adding labor-productivity heterogeneity restores a role for the wealth distribution through two covariances. The belief-invariance result is related to block-recursive equilibria (Menzio and Shi, 2010), but arises from preference structure rather than from a search environment. The quantitative mechanism connects to evidence on heterogeneous returns (Fagereng et al., 2020; Bach et al., 2020), cyclical earnings risk (Storesletten et al., 2004), and state-dependent macroeconomic dynamics (Bloom et al., 2018; Berger and Vavra, 2015).

The remainder of the paper proceeds as follows. Section 2 sets up the model. Section 3 proves the MRCE and belief-invariance results. Section 4 presents the streamlined quantitative application and validation. Section 5 concludes. Proofs, extensions, and additional applications are collected in the Online Appendix.

2 Model

A continuum of infinitely lived households face idiosyncratic labor productivity and portfolio return risk and choose consumption, saving, and labor supply each period.

2.1 States and shocks

Each household's state is $x = [a, z, e]$: wealth a (endogenous), idiosyncratic labor productivity z (exogenous), and a return-scaling factor e (exogenous). All three display substantial cross-sectional variation (Kuhn et al., 2020; Storesletten et al., 2004; Fagereng et al., 2020).

The idiosyncratic processes follow stationary AR(1) dynamics in levels with unit mean:

$$z' - 1 = \rho_z (z - 1) + \varepsilon'_z, \quad \varepsilon'_z \sim \mathcal{N}(0, \sigma^2_{\varepsilon_z}), \quad \sigma^2_{\varepsilon_z} = \sigma^2_z(1 - \rho^2_z), \tag{1}$$

$$e' - 1 = \rho_e (e - 1) + \varepsilon'_e, \quad \varepsilon'_e \sim \mathcal{N}(0, \sigma^2_{\varepsilon_e}), \quad \sigma^2_{\varepsilon_e} = \sigma^2_e(1 - \rho^2_e), \tag{2}$$

where σ^2_z and σ^2_e denote the unconditional variances and $\varepsilon'_z, \varepsilon'_e$ are mutually independent innovations. Linearity in levels is essential: it ensures that the cross-sectional covariances $\text{cov}(z, a)$ and $\text{cov}(e, a)$ serve as sufficient statistics for contemporaneous equilibrium allocations (Section 3). In the quantitative analysis, each process is discretized using the Tauchen (1986) method.

Each household's effective labor productivity per hour is $Z_L z$, where Z_L is an aggregate labor productivity parameter and z is the idiosyncratic component ($\mathbb{E}[z] = 1$). Z_L captures aggregate labor market conditions—utilization intensity, matching quality, and workforce composition—that scale effective household earnings without entering the aggregate production function. This asymmetry makes Z_L a structural shock distinct from TFP (see Section 4). At steady state, $Z_L = 1$.

The aggregate state is $X = [\Phi, A, Z_L, \beta, B, C]$: the cross-sectional distribution Φ, aggregate TFP A, aggregate labor productivity Z_L, the discount factor β, return dispersion B, and productivity dispersion C. In the quantitative application (Section 4), A, Z_L, and β vary; B and C are held at steady state. The framework accommodates time-variation in all five—e.g., widening return dispersion during financial crises or shifting patience during demand-driven episodes.

2.2 Preferences

Household preferences take a Stone-Geary form with endogenous labor supply:

$$u(c, n, a, z; X) = \log(c - g(x; X)) + \theta \log((\kappa + \xi a) - n), \tag{3}$$

where c is consumption, n is labor supply, and $\theta > 0$ governs the relative weight on leisure.

The subsistence level $g(x; X) = \kappa w(X)(1 - \tau) Z_L z$ captures work-related expenses that

scale with after-tax earnings.

The effective time endowment $\kappa+\xi a$ increases in wealth: $\kappa \in (0,1)$ is the base endowment, and $\xi > 0$ captures wealth buying usable time through time-saving services (Aguiar and Hurst, 2007). This wealth-leisure complementarity links the wealth distribution to aggregate labor supply.

The parameters (θ, κ, ξ) are calibrated to match average hours, the saving-to-labor-income ratio, and the level of wealth concentration.

2.3 Budget constraint and the household problem

The recursive formulation of the household's problem is:

$$v(x;X) = \max_{c,a',n} \quad u(c,n,a,z;X) + \beta \mathbb{E} v(x';X') \tag{4}$$

$$\text{s.t.} \quad c + a' = w(X)(1-\tau)Z_L z n + (1 + r(X)(1-\tau)e)a \tag{5}$$

$$\Phi' = \Gamma_\Phi(X) \tag{6}$$

$$a' \geq 0, \quad n \geq 0 \tag{7}$$

The household's problem does not include the transfer T: households are unaware of its existence and optimize based solely on factor income. The actual law of motion for wealth is $a'_{\text{actual}} = a'_{\text{chosen}} + T(X)$, where T is added by the government after the household's decision (see Section 2.4). The wage $w(X)$ is per hour of labor and $r(X)$ is the aggregate return on capital. The factor e scales returns: $\mathbb{E}[e] = 1$, so $r(X)$ is the economy-wide average. The product $Z_L z$ determines the household's effective hourly wage; Z_L captures aggregate labor market conditions while z captures individual heterogeneity. The discount factor $\beta \in (0,1)$ is an element of the aggregate state X and may vary over time, capturing shifts in patience due to financial conditions, uncertainty, or other demand-side forces. The household faces a borrowing constraint $a' \geq 0$, which is non-binding in equilibrium (shown below).

2.4 Production

A representative firm operates a constant-returns-to-scale production technology:

$$Y = AK_{eff}^{\alpha} L^{1-\alpha}, \tag{8}$$

where the inputs are *efficiency-weighted* factor aggregates:

$$K_{eff} \equiv \int e\, a\, d\Phi = K + \mathrm{cov}(e, a), \qquad L \equiv Z_L \int z\, n(x; X)\, d\Phi, \tag{9}$$

with $K = \int a\, d\Phi$ denoting physical (wealth) capital and $\alpha \in (0, 1)$. The return factor e_i scales the efficiency of household i's capital in production (e.g., reflecting portfolio quality or entrepreneurial ability), just as z_i scales labor productivity. Factor prices satisfy:

$$r(X) = \alpha A K_{eff}^{\alpha-1} L^{1-\alpha} - \delta, \quad w(X) = (1-\alpha) A K_{eff}^{\alpha} L^{-\alpha}, \tag{10}$$

where δ is the depreciation rate. Competitive factor markets pay $(r + \delta)$ per efficiency unit of capital and w per efficiency unit of labor. Household i supplies $e_i a_i$ efficiency units of capital and $Z_L z_i n_i$ efficiency units of labor, earning $(r+\delta)e_i a_i - \delta a_i$ and $w Z_L z_i n_i$ respectively. Aggregating: total capital income $(r+\delta)K_{eff} = \alpha Y$ and total labor income $wL = (1-\alpha)Y$, exhausting output. Because factor prices depend on K_{eff}/L, distributional forces—both labor supply shifts and return-wealth sorting—feed back into prices.

2.5 Government policy

The government levies proportional income taxes—at rate τ_K on capital income and τ_L on labor income—and redistributes a fraction η of tax revenue as a lump-sum transfer to all households:

$$T(X) = \eta \cdot \big[\tau_K (r + \delta) K_{eff} + \tau_L w L\big], \tag{11}$$

When $\tau_K = \tau_L = \tau$, this reduces to $T = \eta\tau Y$ (since $(r+\delta)K_{eff} + wL = Y$). After-tax prices are $\tilde{r} = r(1-\tau_K)$ and $\tilde{w} = w(1-\tau_L)$.[1] Remaining revenue finances government consumption (not in utility or production). The baseline sets $\tau_K = \tau_L = 0.30$; the Online Appendix considers time-varying rates.

The transfer plays a dual role. First, it is a fiscal policy channel: tax revenue is partially redistributed, with η controlling the degree of redistribution. Higher η compresses the wealth distribution and thins the Pareto tail. Second, it provides the additive intercept in the Kesten process governing wealth dynamics, which is necessary for a stationary distribution with a Pareto tail to exist (Kesten, 1973). Without $T > 0$, the purely multiplicative system $a' = \beta R_i a$ collapses to the degenerate distribution at zero and admits no non-degenerate stationary distribution.

Timing. The transfer arrives after the household's consumption-saving decision: households solve their problem without knowledge of T, and the government adds T to wealth at the end of the period. This timing is essential for preserving the analytical tractability. If households anticipated T, they would incorporate its present value into effective wealth, breaking the $v = A \log a + B$ value function conjecture and the multiplicative-additive separation $a' = \beta R_i a + T$ that enables closed-form aggregation.

2.6 Recursive competitive equilibrium

Definition 1 (Recursive competitive equilibrium (RCE))**.**
A collection $(g_c, g_a, g_n, v, \Gamma_\Phi, w, r)$ is a recursive competitive equilibrium if:

1. *g_c, g_a, g_n, v solve the household's problem given Γ_Φ.*
2. *K, L satisfy the production sector's optimality conditions.*
3. *Factor markets clear.*

[1] The transfer formula (11) taxes gross capital income $(r+\delta)K_{eff}$, while the household's after-tax return $\tilde{r} = r(1-\tau_K)$ is defined on the net return r. This simplification keeps the household problem tractable; the quantitative difference is $\tau_K \delta \approx 1.5$ percentage points at baseline values.

4. *Γ_Φ satisfies $\Gamma_\Phi(X) = \Phi'$, where Φ' is the future distribution of individual states constructed by applying the intertemporal policy function g_a and the exogenous idiosyncratic state processes to Φ.*

The difficulty is that Φ is infinite-dimensional. The next section shows that optimal policies depend on Φ only through three moments—aggregate capital and two covariances—that serve as sufficient statistics for all contemporaneous equilibrium allocations.

3 Moment-Recursive Competitive Equilibrium

3.1 Analytical household policies

The household's optimal policies have explicit solutions that reduce the infinite-dimensional distribution to a finite moment vector.

Proposition 1 (Analytical household policies and wealth dynamics)**.**
The optimal consumption and labor supply policies are:

$$c(x;X) = \frac{1-\beta}{1+\theta}(1 + r(X)(1-\tau)e)a + \left(\frac{1-\beta}{1+\theta}a\xi + \kappa\right) w(X)(1-\tau)Z_L z \tag{12}$$

$$n(x;X) = \max\left\{\kappa(1+\theta) + \xi a - \frac{\theta\, c(x;X)}{w(X)(1-\tau)Z_L z},\ 0\right\} \tag{13}$$

The household's chosen saving is $\hat{a}'(x;X) = \beta a\left((1 + r(X)(1-\tau)e) + w(X)(1-\tau)Z_L z\xi\right)$. The actual law of motion for wealth, after the unanticipated government transfer, is:

$$a'(x;X) = \hat{a}'(x;X) + T(X) = \beta a\left((1 + r(X)(1-\tau)e) + w(X)(1-\tau)Z_L z\xi\right) + T(X). \tag{14}$$

Proof. The budget constraint can be rewritten as:

$$(1+\theta)(c - \kappa Z_L z w(X)(1-\tau)) + a' = \left((1 + r(X)(1-\tau)e) + w(X)(1-\tau)Z_L z\xi\right) a. \tag{15}$$

Under log utility, the standard result of Brock and Mirman (1972) yields $a' = \beta a R_i$. The government then adds the unanticipated transfer: $a'_{\text{actual}} = \beta a R_i + T(X)$. ■

Consumption is linear in wealth with source-dependent MPCs. The saving function $a' = \beta a(\cdot) + T(X)$ separates into a multiplicative component (enabling aggregation) and an additive transfer (ensuring stationarity).

3.2 The saving rate

The saving rate has a closed-form characterization.

Proposition 2 (Analytical saving rate)**.**
The saving rate $s(x;X) = \frac{a'(x;X)-a}{y(x;X)}$*, where* $y(x;X) = w(X)(1-\tau)Z_L z n(x;X) + r(X)(1-\tau)ea + T(X)$ *is after-tax income plus transfer, satisfies:*

$$s(x;X) = \frac{a[\beta(1+\tilde{r}\,e+\tilde{w}\,Z_L z\xi)-1]+T}{\tilde{w}\,Z_L z\,\kappa\theta + a\left[\frac{\tilde{w}\,Z_L z\,\xi(1+\theta\beta)}{1+\theta} - \frac{\theta(1-\beta)(1+\tilde{r}\,e)}{1+\theta} + \tilde{r}\,e\right] + T}, \tag{16}$$

where $\tilde{r} = r(X)(1-\tau)$ *and* $\tilde{w} = w(X)(1-\tau)$*, with the following properties:*

(i) $\lim_{a\to\infty} s(x;X) = \dfrac{\beta(1+\tilde{r}\,e+\tilde{w}\,Z_L z\xi)-1}{\frac{\tilde{w}\,Z_L z\,\xi(1+\theta\beta)}{1+\theta} - \frac{\theta(1-\beta)(1+\tilde{r}\,e)}{1+\theta} + \tilde{r}\,e}$

(ii) $\lim_{a\to 0} s(x;X) > 0$ *due to the transfer* $T(X) > 0$

(iii) $\frac{\partial s}{\partial a} > 0$ *and* $\frac{\partial^2 s}{\partial a^2} < 0$.

Proof. See Appendix A.2 of the Online Appendix. ■

The saving rate is positive at zero wealth (due to T), increasing, and concave—the pattern documented by Dynan et al. (2004). Concavity generates mean reversion in relative wealth: as wealth rises, the subsistence floor absorbs a diminishing share and the saving rate converges to a constant.

3.3 Non-binding borrowing constraint

An important consequence of the policy functions is that the borrowing constraint is always slack:

Corollary 1 (Non-binding borrowing constraint)**.**
The borrowing limit $a' \geq 0$ *is always non-binding.*

Proof. Given $a'(x;X) = \beta a((1 + r(X)(1-\tau)e) + w(X)(1-\tau)Z_L z\xi) + T(X) \geq T(X) > 0$, the result is immediate. ■

Non-binding constraints keep policies globally linear, enabling analytical aggregation. This is a scope condition: the MRCE applies when transfers are large enough relative to borrowing needs that the constraint never binds. Where binding constraints create kinks, the analytical aggregation breaks down. The condition is broadly consistent with the U.S. economy given Social Security, Medicare, and other social insurance.

3.4 Existence and characterization

Definition 2 (Moment-recursive competitive equilibrium (MRCE))**.**
A collection $(g_c, g_a, g_n, v, \Gamma_M, K, L, w, r, \mathcal{M})$ *is a moment-recursive competitive equilibrium if:*

1. g_c, g_a, g_n, v *solve the household's problem given* Γ_M*.*
2. K, L *satisfy the production sector's optimality conditions.*
3. *Factor markets clear.*
4. Γ_M *satisfies* $\Gamma_M(\mathcal{M}(\Phi), A, Z_L, \beta) = \mathcal{M}(\Phi')$*, where* $\mathcal{M}$ *extracts the sufficient moments from* Φ*, and* Φ' *is the distribution of individual states induced by* g_a *and the exogenous processes.*

The key departure from the RCE is that Γ_M maps a *finite* moment vector to contemporaneous equilibrium allocations and prices. The sufficient statistics are aggregate capital K and two covariances: $\text{cov}(z, a)$ measures the alignment between earnings ability and wealth; $\text{cov}(e, a)$ measures the alignment between portfolio returns and wealth.

Proposition 3 (Unique existence of the MRCE)**.**
The MRCE uniquely exists with $\mathcal{M}(\Phi) = [K, cov(z, a), cov(e, a)]$*. Define after-tax prices* $\tilde{r} = r(\mathcal{M}, A)(1-\tau)$*,* $\tilde{w} = w(\mathcal{M}, A)(1-\tau)$*, and the transfer* $T = \eta \cdot Y \cdot \tau$*. Then:*

(i) Contemporaneous sufficiency. Given $\mathcal{M}$ and the aggregate state (A, Z_L, β), all equilibrium allocations—individual consumption $c(x; X)$, saving $a'(x; X)$, labor supply $n(x; X)$, and aggregate prices r, w, L—are fully determined without reference to the distribution Φ beyond its moments $\mathcal{M}$.

(ii) Aggregate capital law of motion. K' has an exact closed-form expression:

$$K' = \beta\left[(1 + \tilde{r} + \tilde{w} Z_L \xi)K + \tilde{w} Z_L \xi \, cov(z, a) + \tilde{r} \, cov(e, a)\right] + T \tag{17}$$

(iii) Factor prices and aggregate labor. Defining $K_{eff} = K + cov(e, a)$:

$$r(\mathcal{M}, A) = \alpha A \left(\frac{L}{K_{eff}}\right)^{1-\alpha} - \delta \tag{18}$$

$$w(\mathcal{M}, A) = (1 - \alpha) A \left(\frac{K_{eff}}{L}\right)^{\alpha} \tag{19}$$

$$L(\mathcal{M}, A, Z_L, \beta) = Z_L \kappa + \frac{Z_L \xi (1 + \theta\beta)(K + cov(z, a))}{1 + \theta} - \frac{\theta(1 - \beta)}{(1 + \theta)\tilde{w}}[K(1 + \tilde{r}) + cov(e, a)\tilde{r}] \tag{20}$$

Proof. See Appendix A.3 of the Online Appendix. ■

The K' equation reveals three distinct channels. The first term $(1 + \tilde{r} + \tilde{w} Z_L \xi)K$ is the representative-agent channel. The second, $\tilde{r} \operatorname{cov}(e, a)$, is the *return-sorting channel*: high-return households are disproportionately wealthy, raising aggregate saving above what average returns and average wealth would predict. The third, $\tilde{w} Z_L \xi \operatorname{cov}(z, a)$, is the *productivity-sorting channel*: productive-wealthy households generate additional saving through wealth-leisure complementarity. These terms make dynamics *state-dependent*: the same shock produces different responses depending on the degree of wealth sorting.

Distributional dynamics and the closure problem. While $\mathcal{M}$ is sufficient for all contemporaneous equilibrium allocations and K' is exactly determined, the forward evolution of $\operatorname{cov}(z', a')$ and $\operatorname{cov}(e', a')$ generally depends on higher-order moments of the joint distribution

Φ. Specifically, from $a' = \beta a\big((1+\tilde{r}e) + \tilde{w}Z_L z\xi\big) + T$ and $z' = \rho_z z + \varepsilon_z$:

$$\operatorname{cov}(z', a') = \rho_z \beta\big[(1+\tilde{r}+\tilde{w}Z_L\xi)\operatorname{cov}(z,a) + \tilde{w}Z_L\xi\,\mathbb{E}[\hat{z}^2 a] + \tilde{r}\,\mathbb{E}[\hat{z}\hat{e}a]\big], \tag{21}$$

where $\hat{z} = z - 1$ and $\hat{e} = e - 1$. The terms $\mathbb{E}[\hat{z}^2 a]$ and $\mathbb{E}[\hat{z}\hat{e}a]$ involve three-way cross-moments that are not functions of $\mathcal{M}$ alone. Under joint normality of $(\hat{z}, \hat{e}, a)$, these reduce to $\mathbb{E}[\hat{z}^2 a] = \sigma_z^2 K$ and $\mathbb{E}[\hat{z}\hat{e}a] = 0$ (by the Isserlis/Wick theorem, using independence of z and e), and the moment dynamics close. However, the Kesten-process wealth dynamics generate Pareto tails that violate normality.

Therefore, tracking the full distribution via agent-based simulation is necessary for the forward evolution of distributional moments, while the exact household policies, the K' equation (17), and all contemporaneous allocations are preserved analytically. The quantitative analysis in Section 4 employs this approach: exact policies govern each agent's behavior, and the distributional moments are computed from the simulated cross-section.

3.5 Ergodicity

Corollary 2 (Kesten stationarity)**.**
The individual law of motion $a' = \beta a\big((1+\tilde{r}e) + \tilde{w}Z_L z\xi\big) + T$ is a Kesten process with multiplicative coefficient $R_i = \beta\big((1+\tilde{r}e) + \tilde{w}Z_L z\xi\big)$ and additive intercept $T > 0$. When $\mathbb{E}[\log R_i] < 0$, the wealth distribution admits a unique stationary solution with Pareto tails (Kesten, 1973). The additive transfer $T > 0$ is essential: without it, the purely multiplicative system admits only the degenerate distribution at zero.

Proof. See Appendix A.4 of the Online Appendix. ■

The near-unit-root property of the dominant eigenvalue (empirically ≈ 1.01 in the calibrated model) generates slow mean reversion of the wealth distribution and amplifies shock propagation, as shown in Section 4.

3.6 Special cases

The MRCE nests several benchmark models, clarifying what each source of heterogeneity contributes to aggregate dynamics.

Corollary 3 (Krusell and Smith (1998) analogue)**.**
Suppose return heterogeneity is muted: $e = 1$. Then, the MRCE uniquely exists with $\mathcal{M}(\Phi) = [K, cov(z, a)]$, and the aggregate capital law of motion is:

$$K' = \beta\left[(1 + \tilde{r} + \tilde{w}Z_L\xi)K + \tilde{w}Z_L\xi\, cov(z, a)\right] + T \tag{22}$$

Proof. See Appendix A.5 of the Online Appendix. ■

Without return heterogeneity, the state reduces to $[K, \mathrm{cov}(z, a)]$—the analytical counterpart of Krusell and Smith (1998). Capital alone suffices if and only if $\mathrm{cov}(z, a) = 0$. Approximate aggregation holds when $\mathrm{cov}(z, a)$ is small relative to K—i.e., when productivity shocks are weakly persistent or wealth mobility is high. The MRCE thus provides exact conditions under which the Krusell and Smith (1998) finding holds as a theorem. Further discussion appears in Appendix B.3.

Corollary 4 (Representative-agent analogue)**.**
Suppose all heterogeneity is muted: $e = 1$ and $z = 1$. Then, the MRCE uniquely exists with $\mathcal{M}(\Phi) = K$, and the law of motion is:

$$K' = \beta\left[(1 + \tilde{r} + \tilde{w}Z_L\xi)K\right] + T, \tag{23}$$

which leads to the aggregate consumption function:

$$C(\mathcal{M}, A, Z_L, \beta) = \frac{1-\beta}{1+\theta}(1 + \tilde{r} + \tilde{w}Z_L\xi)K + \kappa\tilde{w}Z_L. \tag{24}$$

Proof. See Appendix A.5 of the Online Appendix. ■

With all heterogeneity eliminated, both covariances vanish and the model reduces to the stochastic growth model with log preferences and endogenous labor supply (Brock and Mir-

man, 1972). Every departure of the full MRCE from this benchmark is therefore attributable to sorting between wealth and idiosyncratic characteristics.

3.7 Belief invariance

A distinctive feature of the MRCE is invariance to household beliefs about aggregate dynamics, eliminating the fixed-point problem that dominates standard heterogeneous-agent computation.

Definition 3 (Belief-invariant recursive competitive equilibrium (BRCE))**.**
A collection $(g_c, g_a, g_n, v, \Gamma_\Phi, K, L, w, r)$ *is a belief-invariant recursive competitive equilibrium if:*

1. g_c, g_a, g_n, v *solve the household's problem given arbitrary beliefs about aggregate Markov laws of motion for both endogenous and exogenous states.*
2. K, L *satisfy the production sector's optimality conditions.*
3. *Factor markets clear.*
4. Γ_Φ *satisfies* $\Gamma_\Phi(X) = \Phi'$*, where* Φ' *is the future distribution of individual states constructed by applying the policy function* g_a *and the exogenous idiosyncratic process to* Φ*.*

Theorem 1 (Belief invariance)**.**
In the BRCE, the law of motion Γ_Φ *is unaffected by households' expectations about future aggregate states.*

Proof. See Appendix A.6 of the Online Appendix. ■

The saving policy $a'(x; X) = \beta a((1 + r(X)(1-\tau)e) + w(X)(1-\tau)Z_L z\xi) + T(X)$ depends only on current prices and Z_L—not on forecasts. Under log utility, income and substitution effects of future return changes exactly offset, so all policies are belief-invariant and the aggregate law of motion inherits this property.

Proposition 4 (Unique existence of the BRCE)**.**
The BRCE uniquely exists and the allocations are identical to the MRCE.

Proof. See Appendix A.7 of the Online Appendix. ■

Because the MRCE and BRCE coincide, the equilibrium is computed by forward simulation from any initial distribution Φ_0 and aggregate sequences $\{A_t, Z_{L,t}, \beta_t\}$—no fixed-point iteration, no assumption on expectations, no stationarity requirement. Each period, the exact household policies determine individual decisions, aggregate moments are computed from the cross-section, and the distribution evolves forward. The same procedure applies whether aggregates follow stationary processes, deterministic trends, or both. This also enables the exact inversion: structural shocks can be recovered from data without taking a stand on expectation formation.

Remark 1 (Saving share under time-varying β)**.**
Since β is an element of the aggregate state X, it may vary over time. With time-varying β, the value function takes the form $v(a, z, e; X) = \zeta \log a + B(z, e, X)$, where ζ satisfies $\zeta = (1 + \theta) + \beta\, \mathbb{E}[\zeta']$. Under belief invariance, households treat the current β as permanent, yielding $\zeta = (1 + \theta)/(1 - \beta)$ and saving share $s = \beta$. All propositions above hold with the current β from X: the saving policy is $a'(x; X) = \beta\, a\big((1+\tilde{r}e)+\tilde{w}Z_L\xi z\big)+T$, and the aggregate capital law of motion is $K' = \beta\big[(1 + \tilde{r} + \tilde{w}Z_L\xi)K + \tilde{w}Z_L\xi\, cov(z, a) + \tilde{r}\, cov(e, a)\big] + T$, with no additional state variables or expectation formation required. See Online Appendix A.8 for the ζ recursion derivation.

3.8 Macroeconomic implications

State-dependent aggregate dynamics. The capital law of motion (17) decomposes into three channels: the representative-agent channel $(1+\tilde{r}+\tilde{w}Z_L\xi)K$, the return-sorting channel $\tilde{r}\,\mathrm{cov}(e, a)$, and the productivity-sorting channel $\tilde{w}Z_L\xi\,\mathrm{cov}(z, a)$. These distributional terms make aggregate dynamics *state-dependent*: the same shock produces different responses depending on the degree of wealth sorting.

Distributional channels. The aggregate consumption function decomposes as:

$$C(\mathcal{M}, A, Z_L, \beta) = \underbrace{\frac{1-\beta}{1+\theta}(1+\tilde{r}+\tilde{w}Z_L\xi)K + \kappa\tilde{w}Z_L}_{\text{Representative-agent level}} + \underbrace{\frac{1-\beta}{1+\theta}\left(\tilde{r}\,\text{cov}(e,a) + \tilde{w}Z_L\xi\,\text{cov}(z,a)\right)}_{\text{Distributional channel}} \tag{25}$$

The aggregate employment equation reveals a parallel decomposition:

$$L(\mathcal{M}, A, Z_L, \beta) = \kappa + \underbrace{\frac{Z_L\xi(1+\theta\beta)(K+\text{cov}(z,a))}{1+\theta}}_{\text{Leisure enhancement}} - \underbrace{\frac{\theta(1-\beta)[K(1+\tilde{r}) + \tilde{r}\ \overbrace{\text{cov}(e,a)}^{\text{Dist. channel}}\ ]}{(1+\theta)\tilde{w}Z_L}}_{\text{Wealth effect}} \tag{26}$$

The wealth-effect term is amplified by $\text{cov}(e,a)$: wealthy households earning high returns consume more leisure, reducing labor supply. In periods of high return-wealth sorting, a positive productivity shock elicits a smaller employment response as the households with the largest wealth gains withdraw the most labor.

State-dependent inequality dynamics. The K' equation (17) generates a "wealth spiral": productivity shocks raise capital, which strengthens return-wealth sorting through $\text{cov}(e,a)$, which raises future capital. Discounting ($\beta < 1$) and the transfer T counterbalance. The persistence of the sorting mechanism—governed by the near-unit-root dynamics of the wealth distribution—controls the speed of concentration (Appendix B.4).

3.9 Asset pricing implications

Individual return heterogeneity, traditionally viewed as diversifiable, affects aggregate prices once it is sorted with wealth (Constantinides and Duffie, 1996; Storesletten et al., 2004). Households with persistently high returns accumulate disproportionate wealth, shifting labor supply and equilibrium returns:

$$\frac{\partial r(\mathcal{M}, A)}{\partial \text{cov}(e,a)} = \frac{\partial r}{\partial L}\frac{\partial L}{\partial \text{cov}(e,a)} < 0 \tag{27}$$

Higher return-wealth sorting reduces the equilibrium interest rate through two channels: a static one (the wealth effect lowers labor supply, raising the capital-labor ratio) and a

dynamic one (persistent sorting accelerates capital accumulation, depressing future returns). The full decomposition and quantitative implications appear in Appendix C.

4 Quantitative Analysis

The quantitative exercise uses the analytical structure for two purposes. First, it asks whether the inversion recovers economically interpretable shocks rather than simply relabeling three observables. Second, it measures the distributional terms that the theory identifies as missing from representative-agent aggregation. It does not use exact fit as evidence: with three observables and three recovered aggregate shocks, output, capital, and hours are matched mechanically. The tests are instead non-targeted: simulated Krusell–Smith economies, external labor and saving indicators, forecast performance, and distributional implications.

4.1 Inversion and Calibration

The application uses annual U.S. data from 1950 to 2019 on real per-capita GDP, the per-capita capital stock, and aggregate hours. The aggregate equations are the production function, labor supply, and capital accumulation:

$$Y_t = A_t K_{eff,t}^{\alpha} L_t^{1-\alpha}, \tag{28}$$

$$L_t = Z_{L,t}\kappa + \frac{Z_{L,t}\,\xi(1+\theta\beta_t)}{1+\theta}(K_t + \mathrm{cov}_t(z,a)) - \frac{\theta(1-\beta_t)}{(1+\theta)\tilde{w}_t}\big(K_t(1+\tilde{r}_t) + \tilde{r}_t\,\mathrm{cov}_t(e,a)\big), \tag{29}$$

$$K_{t+1} = \beta_t\left[(1+\tilde{r}_t + \tilde{w}_t Z_{L,t}\xi)K_t + \tilde{r}_t\,\mathrm{cov}_t(e,a) + \tilde{w}_t Z_{L,t}\xi\,\mathrm{cov}_t(z,a)\right] + T_t. \tag{30}$$

Note that equation (29) is implicit in L because prices depend on L through the marginal products (Online Appendix A proves unique existence). The distributional covariances entering $K_{eff,t}$ are obtained from a preliminary two-series inversion that provides the moment-recursion covariance path (Online Appendix E). Given the distributional state, these equations recover $(A_t, \beta_t, Z_{L,t})$ period by period. TFP is identified from production, the saving-

margin wedge β_t from capital accumulation, and the labor-efficiency wedge $Z_{L,t}$ from labor supply. The distribution is then simulated forward using the exact household policy.

The calibration is deliberately conventional. The capital share is $\alpha = 0.33$; depreciation is time-varying from BEA fixed-assets data; the time-varying capital share α_t from Penn World Tables is used in the inversion (the analytical expressions carry over with α replaced by α_t period by period, since each period's equilibrium is contemporaneous); $\beta = 0.96$ is the reference annual discount factor; the leisure weight is $\theta = 4$ (Online Appendix E.12 reports robustness to alternative values); idiosyncratic earnings risk follows Storesletten et al. (2004); and idiosyncratic return risk follows Fagereng et al. (2020). The transfer share is calibrated to place steady-state wealth concentration near the U.S. range.

Table 1: Calibrated parameters

Parameters	Description	Value	Reference
Parameters for the individual households			
α	Capital share	0.3300	
β	Discount factor	0.9600	
δ_t	Depreciation rate (time-varying)	0.037–0.058	BEA Fixed Assets
θ	Labor disutility	4.0000	KS98 validation
κ	Common time endowment	0.2700	$\kappa < \min(L_{\text{data}})$
ξ	Wealth–leisure complementarity	0.0213	Saving/Labor Income ≈ 0.33
ρ_z	Idiosyncratic prod. persistence	0.9500	Storesletten et al. (2004)
ρ_e	Idiosyncratic return persistence	0.9990	Fagereng et al. (2020)
σ_z	Idiosyncratic prod. volatility	0.1200	Storesletten et al. (2004)
σ_e	Idiosyncratic return volatility	0.0100	Fagereng et al. (2020)
Government policy parameters			
τ	Proportional income tax rate	0.3000	Effective tax rate
η	Transfer share of tax revenue	0.0133	Wealth Gini ≈ 0.80
Parameters for the business cycle			
ρ_A	Aggregate TFP persistence	0.9500	Cooley and Prescott (1995)
ρ_{σ_e}	Uncertainty shock persistence	0.7800	Bloom et al. (2018)
σ_A	Aggregate TFP volatility	0.0070	Cooley and Prescott (1995)
σ_{σ_e}	Uncertainty shock volatility	0.3600	Bloom et al. (2018)

Notes: Calibration is annual. The table reports the baseline used in the U.S. inversion and in the distributional simulation.

4.2 Validation and Recovered Shocks

Krusell–Smith validation. The first check is whether the inversion spuriously creates additional shocks when the data are generated by a model with only aggregate TFP risk. I simulate three Krusell–Smith variants and apply the same inversion to simulated output, capital, and hours. Figure 1 and Table 2 show that the recovered TFP matches the true process, while β_t and $Z_{L,t}$ remain essentially constant. Two extensions, one with endogenous labor supply and one with Frisch labor supply and convex adjustment costs, deliver the same result. The inversion therefore does not mechanically manufacture preference or labor shocks when the true data-generating process has only aggregate TFP.

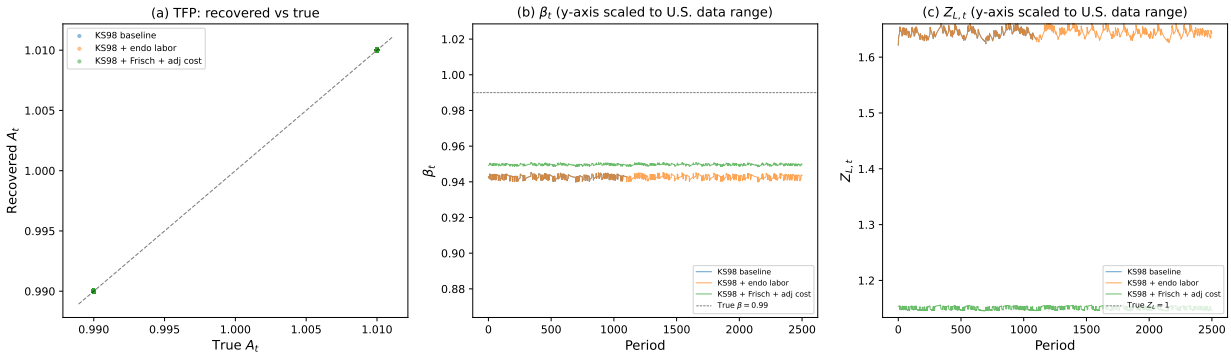

Figure 1: Krusell–Smith validation: recovered shocks from simulated data

Notes: Sequential inversion applied to simulated (Y_t, K_t, L_t) from three Krusell–Smith variants. The recovered TFP aligns with the true aggregate TFP process, while the recovered β_t and $Z_{L,t}$ series are nearly flat on the U.S. data scale.

Table 2: Krusell–Smith validation: inversion accuracy

	$\text{mean}\lvert A^{\text{inv}} - A^{\text{true}}\rvert/A$	$\text{CV}(\beta)$	$\sigma(\Delta \log Z_L)$	$\text{mean}\lvert Y^{\text{inv}} - Y^{\text{true}}\rvert/Y$
KS98 baseline	7.5×10^{-6}	0.12%	0.17%	$< 10^{-15}$
KS98 + endo labor	7.3×10^{-6}	0.12%	0.17%	$< 10^{-15}$
KS98 + Frisch + adj cost	2.5×10^{-5}	0.05%	0.18%	$< 10^{-15}$
U.S. data (1950–2019)	—	1.78%	3.42%	exact

Notes: In the simulated economies, β and Z_L are constant by construction and only aggregate TFP varies. The final row applies the same inversion to U.S. data.

Applying the same inversion to U.S. data reveals economically significant departures from the Krusell–Smith benchmark. The discount factor β_t fluctuates more than in the

simulated economies, but the labor-efficiency wedge is the dominant departure. U.S. capital accumulation is well approximated by a single TFP shock with near-constant preferences, while labor-market fluctuations require an additional structural margin that a one-shock Krusell–Smith economy cannot generate.

Recovered series. Figure 2 reports the recovered structural series. TFP grows from 0.802 in 1950 to 1.78 in 2019, tracing the postwar expansion, the 1970s slowdown, the IT revolution, and post-2008 stagnation. The saving-margin wedge β_t is persistent and countercyclical, with a secular decline from 1950 to 2019. The labor-efficiency wedge $Z_{L,t}$ fluctuates at business-cycle frequencies and captures movements in effective labor input beyond measured hours.

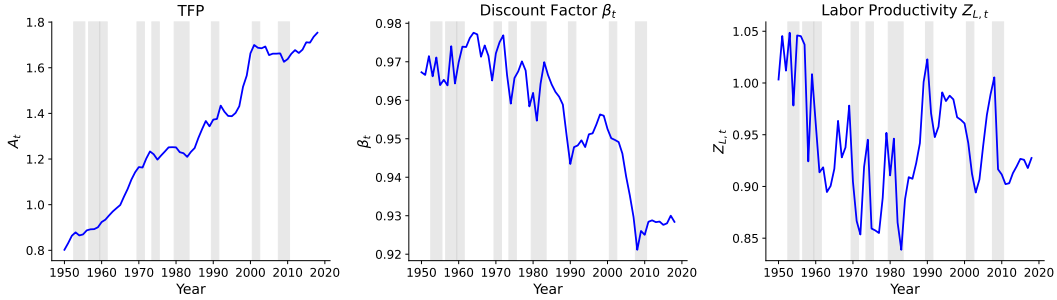

Figure 2: Recovered structural series from the data-anchored inversion

Notes: Top: aggregate TFP. Middle: saving-margin wedge β_t. Bottom: labor-efficiency wedge $Z_{L,t}$. The inversion uses observed output, capital, and hours. Shaded areas indicate NBER recession dates.

External labor discipline. The second check asks whether the recovered labor-efficiency wedge is related to external labor-market margins not used in the inversion. At business-cycle frequencies, $Z_{L,t}$ is strongly correlated with the employment-population ratio and labor-force participation, the extensive margins it is designed to capture. The BLS labor composition index, by contrast, has the opposite cyclical sign: in recessions, low-productivity workers are disproportionately laid off, raising measured average quality, while $Z_{L,t}$ falls because utilization and matching deteriorate. The sign reversal supports the interpretation of Z_L as a utilization and matching margin rather than a demographic composition index.

Saving-margin discipline. The declining trend in β_t is not used as a target, so it needs external discipline. Figure 3 compares the recovered saving-margin wedge with three outside

Table 3: External validation: recovered $Z_{L,t}$ and labor-market indicators

External series	Level corr	HP-cycle corr
Employment-population ratio (EMRATIO)	0.22	0.80
Labor force participation rate (CIVPART)	0.01	0.53
Avg. weekly hours, manufacturing (AWHMAN)	0.22	0.41

Notes: The external series are not used in the inversion. HP-cycle correlations use annual data with $\lambda = 6.25$.

series: the personal saving rate, labor-force participation, and revolving consumer credit relative to GDP. The correlations are not a structural proof, but they help interpret β_t as a compact saving-side wedge that absorbs forces operating through the saving margin, including demographic shifts, financial liberalization, and precautionary motives.

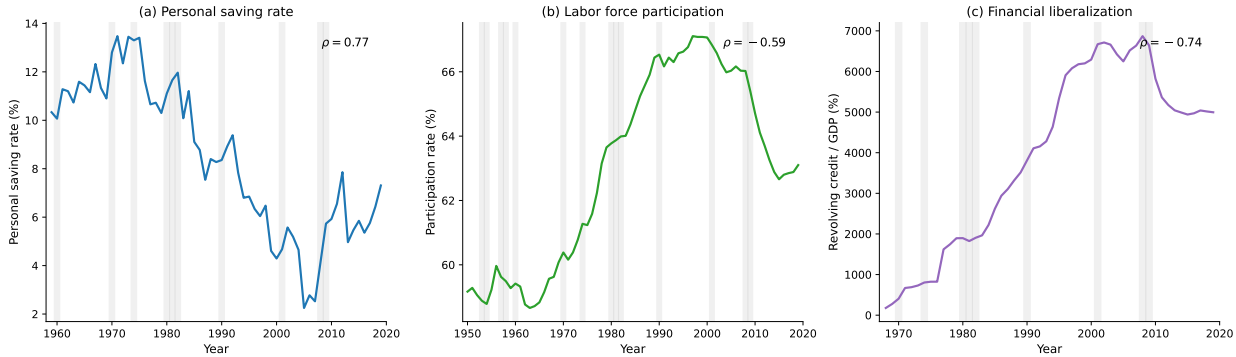

Figure 3: External discipline for the saving-margin wedge

Notes: External indicators from FRED, not used in the inversion. Panels show the U.S. personal saving rate, civilian labor-force participation, and revolving consumer credit as a share of GDP against the recovered β_t. Shaded areas indicate NBER recession dates.

A two-dimensional business cycle. Figure 4 displays pairwise scatter plots of the recovered shocks. The most striking finding is the near-orthogonality of TFP with both $Z_{L,t}$ and β_t: technology evolves independently of the labor-saving block. The key cyclical correlation is between labor efficiency and patience, $\rho(\Delta \log Z_{L,t}, \Delta\beta_t) = -0.884$. Labor-market deterioration is systematically accompanied by higher saving, consistent with a precautionary channel. Together with the Krusell–Smith validation, the scatter plot shows that the inversion is not merely relabeling one residual: trend productivity and the labor-saving block occupy different dimensions of the data.

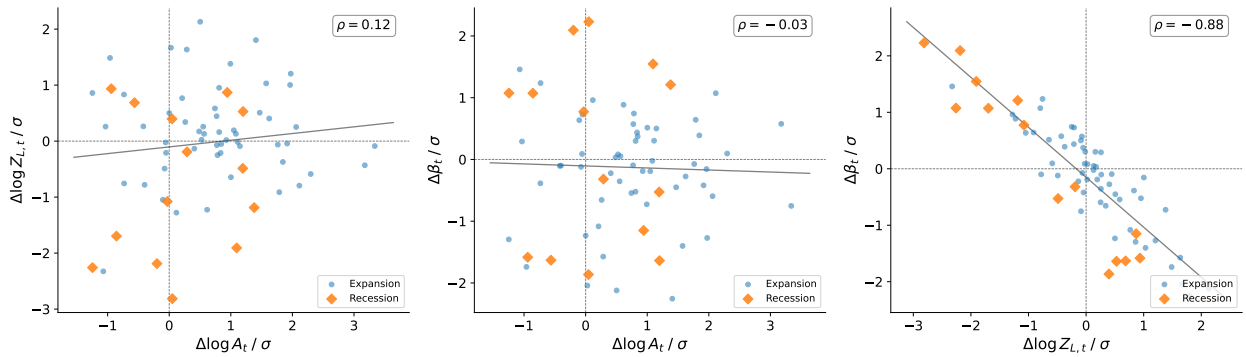

Figure 4: Pairwise scatter of recovered shock changes: the two-dimensional business cycle

Notes: Each point is one year. Blue circles are expansion years; orange diamonds are recession years. Changes are normalized by their unconditional standard deviations. The right panel shows the labor-saving tandem; the left and center panels show that TFP innovations are nearly orthogonal to the other two shocks.

4.3 Conditional Forecasting

A distinctive advantage of the MRCE is that conditional forecasting is computationally direct. In conventional heterogeneous-agent models, forecasting requires re-solving the household dynamic program or estimating a perceived aggregate law of motion, because policy functions depend on beliefs about future aggregate states. In the MRCE, household policies are backward-looking and linear in wealth. Given any future sequence for $(A_t, \beta_t, Z_{L,t})$, the model produces output, capital, labor, prices, consumption, investment, and the distributional state by forward simulation alone.

Rolling forecasts. I exploit this property with a rolling forecast exercise. For each origin year $t \in \{1965, \ldots, 2009\}$, the inversion recovers structural shocks from data available up to t, and the model is simulated forward holding shocks at their last recovered values. The resulting forecast jointly produces output, capital, labor, prices, consumption, investment, and the distributional state, all satisfying the production function, market clearing, and the resource constraint simultaneously.

Figure 5 compares the MRCE forecast to a recursive AR benchmark. The benchmark forecasts the structural inputs using the same information set and aggregates them through the same production function. At both one- and three-year horizons, the MRCE has lower RMSE for all five macro aggregates: output (3.2% vs. 3.9% at $h = 1$; 5.2% vs. 8.0% at $h = 3$), capital (2.2% vs. 15.7%; 2.9% vs. 16.7%), labor (2.8% vs. 4.2%; 5.7% vs. 6.0%),

consumption (4.5% vs. 5.8%; 6.8% vs. 9.5%), and investment (10.6% vs. 11.0%; 12.8% vs. 16.0%). The capital-stock improvement is the most striking (RMSE ratios of 0.14 and 0.17). The recursive AR forecasts drift because each input evolves independently; the MRCE's GE discipline prevents this divergence.

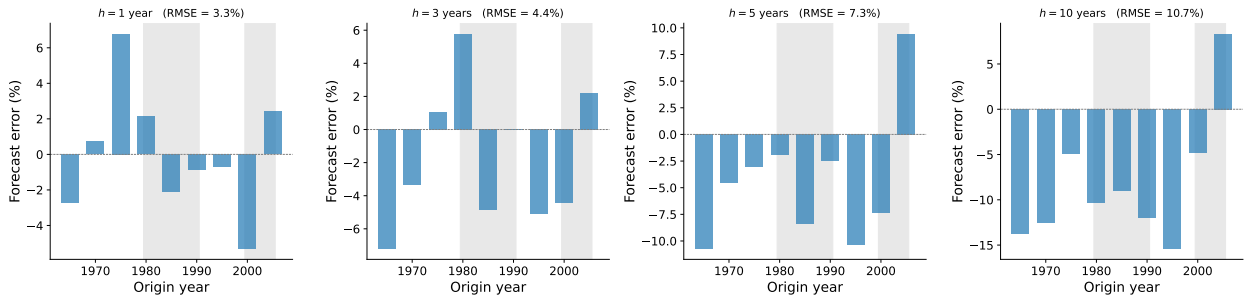

Figure 5: Rolling forecast errors by horizon

Notes: For each origin year, the model is simulated forward holding recovered shocks at their last values. The recursive AR benchmark forecasts structural inputs using the same information set and aggregates through the same production function. Panel titles report RMSEs. Shaded areas indicate NBER recession dates.

4.4 Aggregate Dynamics

The analytical aggregate equations imply three quantitative lessons. First, representative-agent aggregation is an excellent approximation in the baseline calibration. Investment decomposes into a representative-agent channel, a return-sorting channel, and a productivity-sorting channel; the representative-agent channel accounts for 92.2 percent of mean investment. This explains why the Krusell–Smith approximation works so well even though the exact state contains additional covariances.

Second, the omitted covariances still create a state-dependent amplification channel. I compute generalized impulse responses to a 2 percent negative TFP shock from the 1952 distribution, when wealth inequality is low, and from the 2019 distribution, when wealth inequality is high. The impact effect is 3 percent larger from the 2019 state (-2.29% versus -2.22%), and the cumulative 10-year output loss is modestly larger (-20.4% versus -20.2%). The magnitude is small at the conservative baseline calibration, but the sufficient statistic is transparent: amplification rises with return-wealth sorting.

Third, the direct asset-pricing effect of return-wealth sorting is negligible at $\sigma_e = 0.01$.

The interest-rate effect is approximately -0.001 basis points. This is a useful negative result. In the baseline calibration, return heterogeneity matters more as an amplification state variable than as a quantitatively important risk-premium channel. An Epstein–Zin extension, reported in the Online Appendix, is a more promising route for asset-pricing applications.

4.5 Distributional Dynamics and the Sorting Accelerator

TFP growth and inequality. The central distributional mechanism is a sorting accelerator. TFP growth raises returns, persistent return heterogeneity causes high-return households to become increasingly wealthy, and wealth concentration then raises the importance of return-wealth sorting in future dynamics. The relevant covariance, $\mathrm{cov}(e, a)$, is therefore highly persistent: the dominant eigenvalue governing distributional dynamics exceeds one in 100 percent of sample years, with mean 1.02.

Four distributional counterfactuals isolate the mechanism. The baseline simulation uses the historical paths of all recovered shocks. The TFP-only counterfactual keeps only the TFP path. The fixed-A counterfactual removes TFP growth. The all-fixed counterfactual holds all aggregate shocks fixed. Figure 6 reports the results. Both sorting channels grow substantially along the baseline path, but for different reasons. Return-wealth sorting grows through near-unit-root persistence, while productivity-wealth sorting grows through recession-driven bursts in idiosyncratic dispersion. Without TFP growth, return-wealth sorting stagnates; without countercyclical dispersion, productivity sorting loses its recession spikes.

Analytical decomposition. The MRCE makes the sorting accelerator precise. In the return-sorting channel, persistence is governed by $\phi_e \rho_e (1 + \tilde{r})$, which is close to $1/\beta$ because return shocks are highly persistent. In the productivity-sorting channel, persistence is lower, but countercyclical dispersion raises the innovation rate during downturns. Figure 7 decomposes these forces. The secular component comes from persistent return heterogeneity; the cyclical component comes from recession-driven increases in idiosyncratic dispersion.

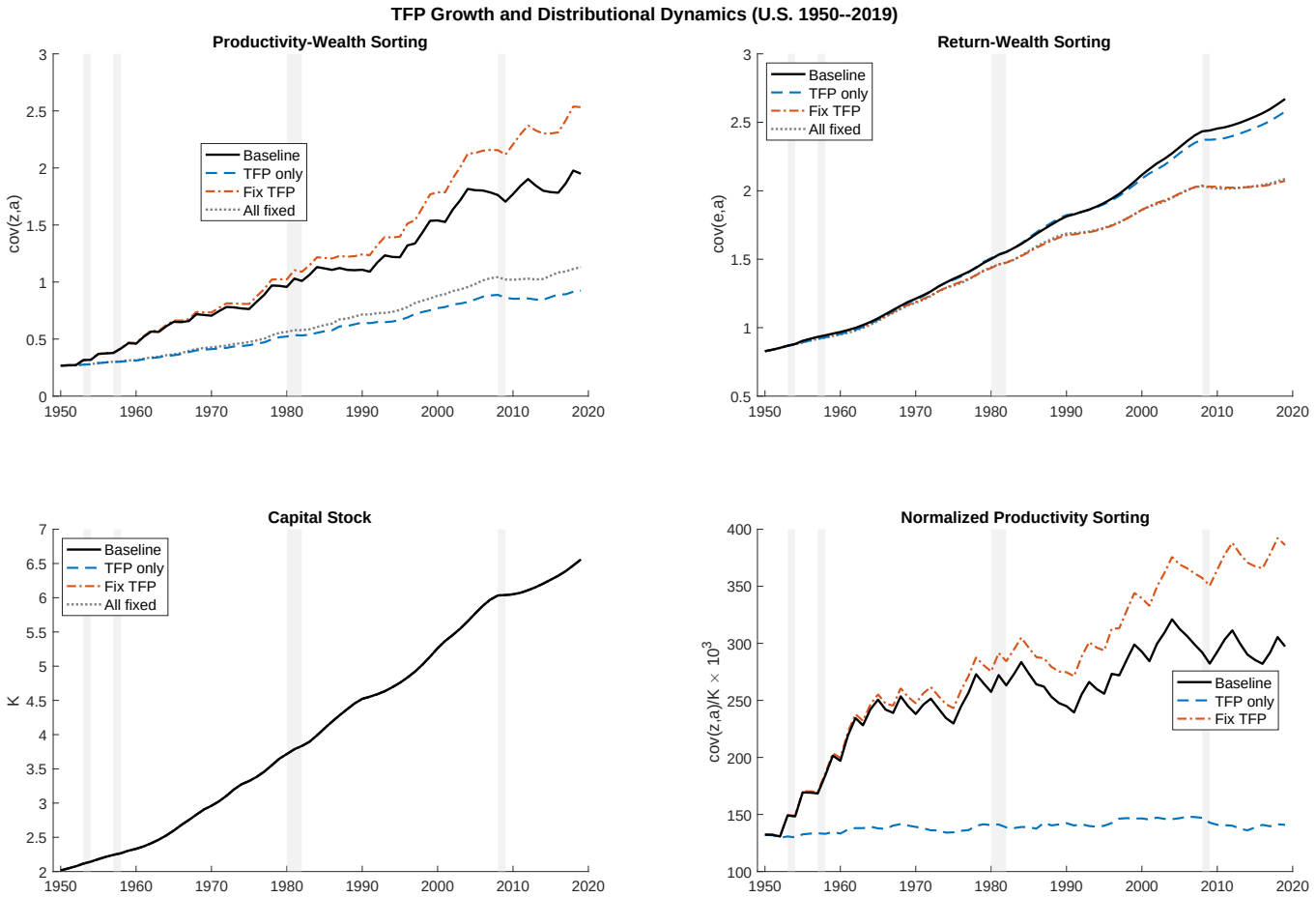

Figure 6: TFP growth and distributional dynamics

Notes: Top left: productivity-wealth covariance $\text{cov}(z, a)$ under four counterfactual scenarios. Top right: return-wealth covariance $\text{cov}(e, a)$. Bottom left: aggregate capital. Bottom right: normalized sorting intensity $\text{cov}(z, a)/K$. Shaded areas indicate NBER recession dates.

Who gains from TFP growth? Table 4 reports the impact of 1 percent TFP growth by wealth percentile. The top decile captures 86.0 percent of the aggregate wealth gain, exceeding its 82.7 percent wealth share, because the return-sorting and productivity-sorting channels compound. The top 1 percent alone captures 72.5 percent. The table is not a tax counterfactual; it is a sufficient-statistic calculation showing how the analytical channels allocate a common productivity improvement across the wealth distribution.

Historical gain shares. The steady-state calculation is conservative because it holds the distribution fixed. Along the historical path, the distribution itself changes. Table 5 reports decade averages of TFP-driven wealth gains by wealth decile, and Figure 8 visualizes the

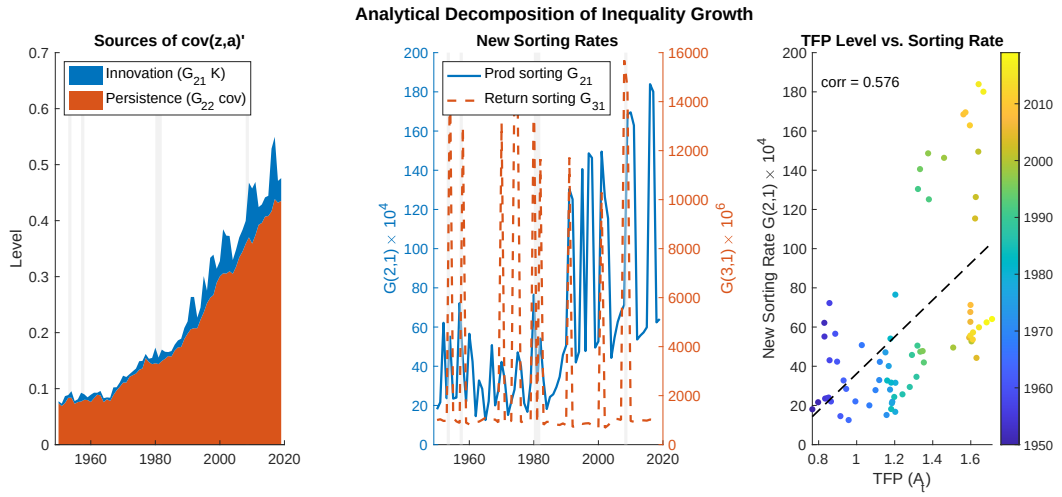

Figure 7: Analytical decomposition of distributional sorting

Notes: Left: innovation and persistence contributions to $\text{cov}(z, a)'$. Center: sorting-rate time paths for productivity and return shocks. Right: TFP level against productivity sorting. Shaded areas indicate NBER recession dates.

Table 4: Distributional impact of 1% TFP growth at steady state

Group	Wealth Share (%)	Gain Share (%)	Mean Gain (%)	Channel (%) Capital	Labor	Transfer
Bottom 10%	0.8	0.6	0.0829	64.4	18.2	17.4
10–20%	1.0	0.8	0.0870	67.5	19.2	13.4
20–30%	1.2	0.9	0.0892	69.2	19.6	11.2
30–40%	1.4	1.1	0.0911	70.6	19.8	9.6
40–50%	1.6	1.2	0.0930	71.8	20.0	8.2
50–60%	1.9	1.5	0.0950	73.1	20.1	6.9
60–70%	2.2	1.8	0.0971	74.4	20.1	5.6
70–80%	2.9	2.4	0.0999	75.9	19.9	4.2
80–90%	4.3	3.7	0.1036	77.9	19.4	2.8
Top 10%	82.7	86.0	0.1117	84.4	15.5	0.1
Top 5%	78.9	82.6	0.1152	84.6	15.4	0.1
Top 1%	68.4	72.5	0.1204	84.9	15.1	0.0

Notes: Gains computed from 1% TFP growth at the stationary distribution of simulated households. "Capital" is the gain from higher after-tax capital income; "Labor" is the gain from higher labor income; "Transfer" is the common transfer component.

evolving distribution of gains. The top decile captures approximately 70 percent of TFP-driven wealth gains throughout the sample. The reason is not simply that the top decile begins wealthy; it is that return-wealth sorting and productivity-wealth sorting compound the effect of each productivity improvement through the exact saving rule.

Table 5: Share of TFP-driven wealth gains by wealth decile (%), decade averages

Decade	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	Top 5%	Top 1%
1950s	1.5	1.9	2.2	2.5	2.8	3.2	3.8	4.6	6.3	71.1	66.2	56.0
1960s	1.3	1.7	2.0	2.3	2.7	3.1	3.6	4.5	6.2	72.6	67.7	57.5
1970s	1.2	1.5	1.8	2.1	2.4	2.9	3.4	4.3	6.0	74.4	69.6	59.4
1980s	1.0	1.4	1.7	2.0	2.3	2.7	3.3	4.1	5.9	75.6	70.7	60.2
1990s	0.9	1.3	1.5	1.8	2.2	2.6	3.1	4.0	5.9	76.6	71.7	60.8
2000s	0.8	1.2	1.4	1.7	2.0	2.4	3.0	3.9	5.9	77.5	72.4	60.8
2010s	0.7	1.0	1.3	1.6	1.9	2.3	2.8	3.8	5.8	78.9	73.8	61.7
Full sample	1.1	1.4	1.7	2.0	2.3	2.7	3.3	4.2	6.0	75.3	70.4	59.6

Notes: Each entry reports the average share of aggregate TFP-driven wealth gains captured by the indicated wealth decile. Gains are computed period by period along the historical simulation.

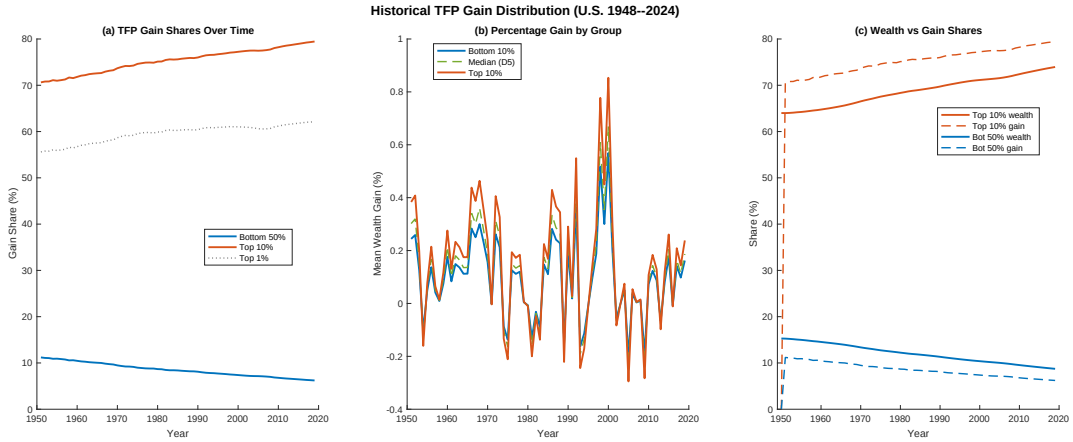

Figure 8: Historical distribution of TFP-driven gains

Notes: Period-by-period decomposition of the share of TFP-driven wealth gains by wealth percentile, U.S. 1950–2019.

4.6 Cross-Sectional Dynamics

The same historical simulation generates cross-sectional moments. Table 6 compares the model's wealth Gini with SCF-based estimates at survey years. The model captures the broad level of U.S. wealth inequality through the Kesten-process rule $a' = \beta R_i a + T$, with persistent return shocks and positive transfers. It rises more smoothly than the data. This is a useful miss: time-varying return dispersion, top income growth, and housing-market divergence are natural candidates for the remaining trend.

Figure 9 summarizes the time-series path of wealth, consumption, and income inequality. Wealth concentration rises gradually along the historical path. Consumption inequality also

rises, because the consumption rule loads on wealth through $R_i a_i$, while income inequality is flatter. This distinction is useful: the model does not try to match every distributional margin, but it does generate a coherent mapping from the exact household policies to cross-sectional consumption and wealth moments.

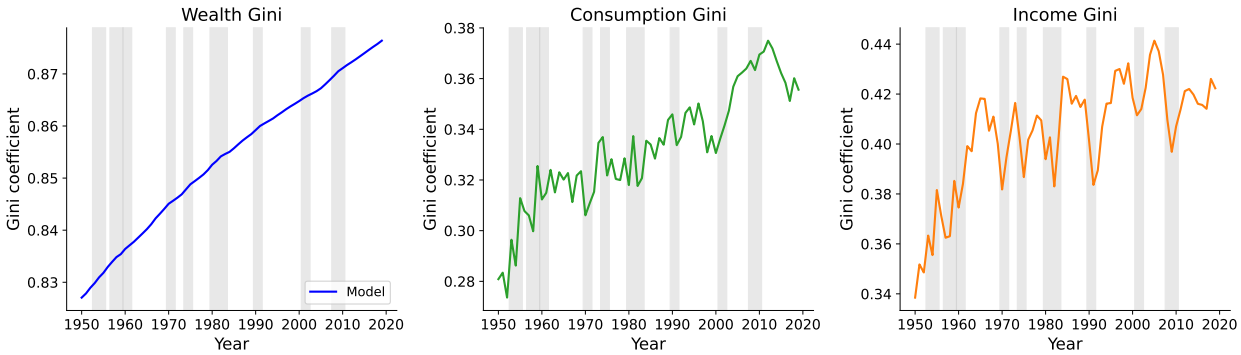

Figure 9: Historical cross-sectional distributional dynamics

Notes: Historical simulation using the recovered aggregate shocks and exact household policies. The wealth panel includes SCF-based estimates as external benchmarks. Shaded areas indicate NBER recession dates.

Table 6: Wealth Gini: model vs. data

Year	Data	Model	Year	Data	Model
1953	0.73	0.830	1998	0.80	0.864
1962	0.78	0.838	2001	0.81	0.865
1968	0.77	0.843	2004	0.81	0.867
1983	0.77	0.855	2007	0.82	0.869
1989	0.79	0.858	2010	0.85	0.871
1992	0.78	0.860	2013	0.85	0.873
1995	0.78	0.862	2016	0.86	0.875

Notes: Data are SCF-based estimates reported by Kuhn and Ríos-Rull (2025). Model moments are from the historical simulation using exact household policies.

The time-series Gini compresses the distribution into one number. Figure 10 overlays the normalized wealth distribution in 1980 and 2015. The upper tail shifts right and the middle thins, illustrating the same sorting force in distributional form: persistent return heterogeneity stretches the wealth distribution even when the aggregate calibration keeps amplification modest.

Table 7 reports consumption and income moments in normal times and across recession episodes. Consumption inequality rises modestly in recessions, while income inequality falls.

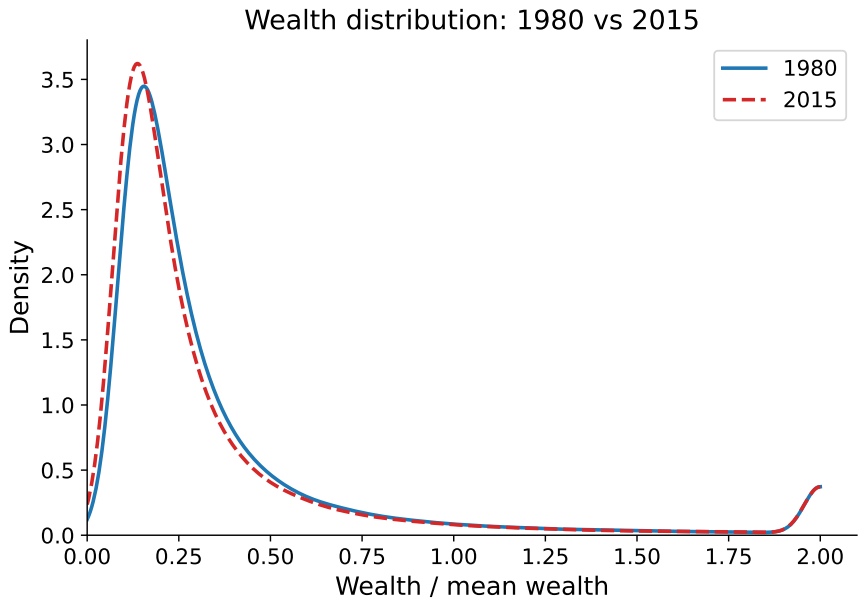

Figure 10: Normalized wealth distribution: 1980 vs. 2015

Notes: Kernel density estimates of wealth divided by mean wealth from the historical household simulation. Values are winsorized at twice mean wealth for readability.

The consumption response reflects the increase in the consumption share and the larger role of wealth-driven heterogeneity when β_t falls; income compression reflects declines in $Z_{L,t}$ and the associated reduction in labor-income dispersion. The Great Recession stands out because high wealth concentration interacts with a sharp fall in the saving-margin wedge, while early postwar recessions show much smaller consumption dispersion.

Consumption dynamics. The MRCE also yields a closed-form aggregate consumption equation,

$$C_t = \frac{1-\beta_t}{1+\theta}\left[(1+\tilde{r}_t+\tilde{w}_t Z_{L,t}\xi_t)K_t + \tilde{r}_t \operatorname{cov}_t(e,a) + \tilde{w}_t Z_{L,t}\xi_t \operatorname{cov}_t(z,a)\right] + \kappa \tilde{w}_t Z_{L,t}.$$

The individual MPC out of a one-time transfer is $(1-\beta_t)/(1+\theta)$, independent of wealth. This is not a HANK-style high-MPC mechanism; the state dependence works through prices, β_t, and the sorting covariances. Figure 11 reports the consumption-output ratio, the annual transfer MPC, and the cumulative steady-state MPC. The low annual MPC is another scope condition: this version of the model is designed to isolate analytical aggregation and sorting,

Table 7: Cross-sectional distributional moments: normal times vs. recessions

	Consumption				Income			
	CV	Gini	Skew	Kurt	CV	Gini	Skew	Kurt
Normal	1.009	0.334	8.19	92.3	1.798	0.408	11.75	178.7
1953–54 (Korean War)	1.003	0.291	10.28	134.2	1.738	0.359	12.89	203.6
1957–58 (Eisenhower)	0.973	0.303	9.42	118.1	1.600	0.363	12.30	189.7
1973–75 (Oil Crisis)	0.897	0.331	7.13	75.9	1.708	0.402	11.94	185.0
1980 (Volcker I)	0.928	0.318	8.15	92.1	1.742	0.394	11.67	167.9
1981–82 (Volcker II)	0.967	0.327	8.16	92.1	1.671	0.393	11.84	182.8
1990–91 (S&L Crisis)	1.168	0.340	9.58	114.8	1.696	0.392	11.99	187.3
2007–09 (Great Recession)	1.276	0.365	9.34	107.4	1.744	0.411	11.80	191.5

Notes: Cross-sectional moments from simulating households along the historical equilibrium path using the three-shock inversion with countercyclical dispersion. CV = coefficient of variation. Normal times are non-NBER-recession years; recession moments are averaged within each episode.

not to reproduce high-frequency liquidity-channel consumption responses.

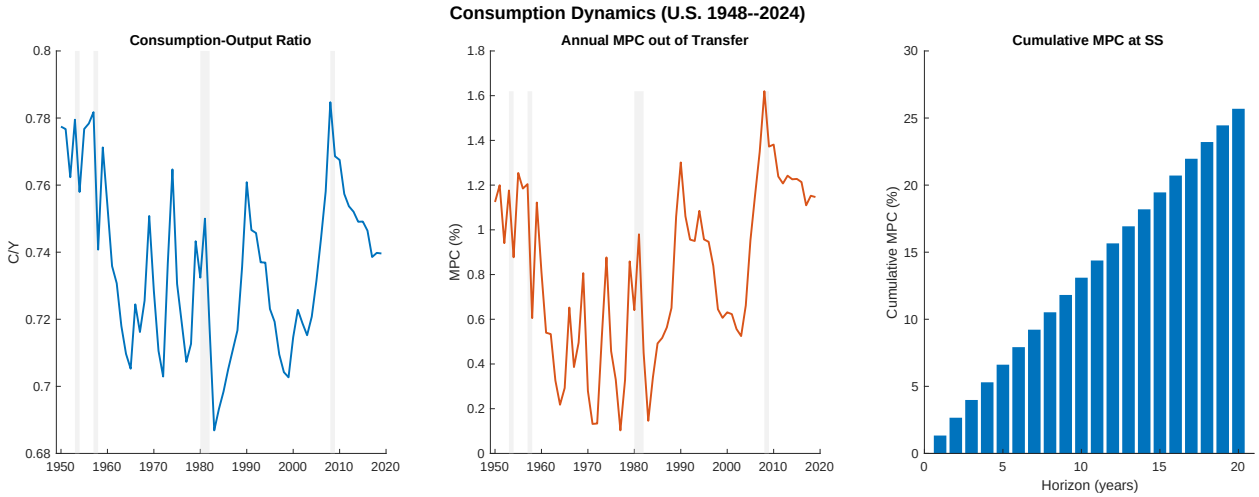

Figure 11: Consumption dynamics and transfer MPC

Notes: Left: consumption-output ratio. Center: annual MPC out of a one-time transfer implied by $(1 - \beta_t)/(1 + \theta)$. Right: cumulative steady-state MPC by horizon.

4.7 Structural recession classification

The exact inversion classifies recessions structurally—by reading off which shock moved—without narrative identification or external instruments. With three shocks, the classification is richer than supply-versus-demand: labor-driven ($Z_{L,t}$ decline), demand-driven (β_t

decline), or TFP-driven (A_t decline). Table 8 reports recovered shocks around all seven NBER recessions in standard-deviation units. The seven recessions divide into three types. The labor-savings tandem (declining Z_L, rising β) drives the 1957–58, 1980, and 1981–82 episodes. The 1973–75 Oil Crisis is demand-driven (β decline). The 2007–09 Great Recession is TFP-driven. This heterogeneity in recession mechanisms illustrates the value of a three-shock framework over simpler decompositions.

Table 8: Structural recession classification from the exact inversion

Episode	ΔY (σ)	$\Delta \log A$ (σ)	$\Delta \log Z_L$ (σ)	$\Delta\beta$ (σ)	Classification
1953–54 (Korean War)	−1.12	−0.86	−1.81	−0.26	Labor-driven
1957–58 (Eisenhower)	−1.16	+0.34	−3.17	+1.77	Labor-driven
1973–75 (Oil Crisis)	−1.27	−1.63	−2.30	−3.04	TFP/Labor
1980 (Volcker I)	−0.68	−0.03	−1.05	+0.61	Labor-driven
1981–82 (Volcker II)	−1.33	−1.14	−2.11	+0.22	TFP/Labor
1990–91 (S&L Crisis)	−0.69	+1.32	−1.22	−1.15	Labor-driven
2007–09 (Great Recession)	−2.07	−1.25	−2.03	−3.27	TFP/Labor

Notes: All entries are peak-to-trough changes in units of each variable's unconditional standard deviation. Classification: "TFP-driven" when $|A|$ dominates; "Labor-driven" when $|Z_L|$ dominates; "Demand-driven" when $|\beta|$ dominates.

4.8 Distributional impact of tax reforms

The near-unit-root persistence of wealth sorting implies that even temporary changes in capital taxation can produce almost permanent shifts in the wealth distribution. I test this prediction using two historically abrupt capital tax changes: Reagan ERTA (1981, τ_K from 0.380 to 0.375) and Bush EGTRRA (2001, τ_K from 0.395 to 0.355).

For each episode, two full agent-based simulations (500,000 agents) use identical aggregate shocks and idiosyncratic histories. The counterfactual freezes $\tau_{K,t}$ at its pre-reform level from the reform year onward; capital and prices evolve endogenously under each scenario. Figure 12 reports the results.

Table 9 summarizes the growth-versus-inequality tradeoff. Each reform raises output and wages but simultaneously increases wealth inequality. The mechanism is return-wealth sorting: lower capital taxes disproportionately raise capital income for households whose wealth is large enough to generate substantial returns. The near-unit-root persistence ensures

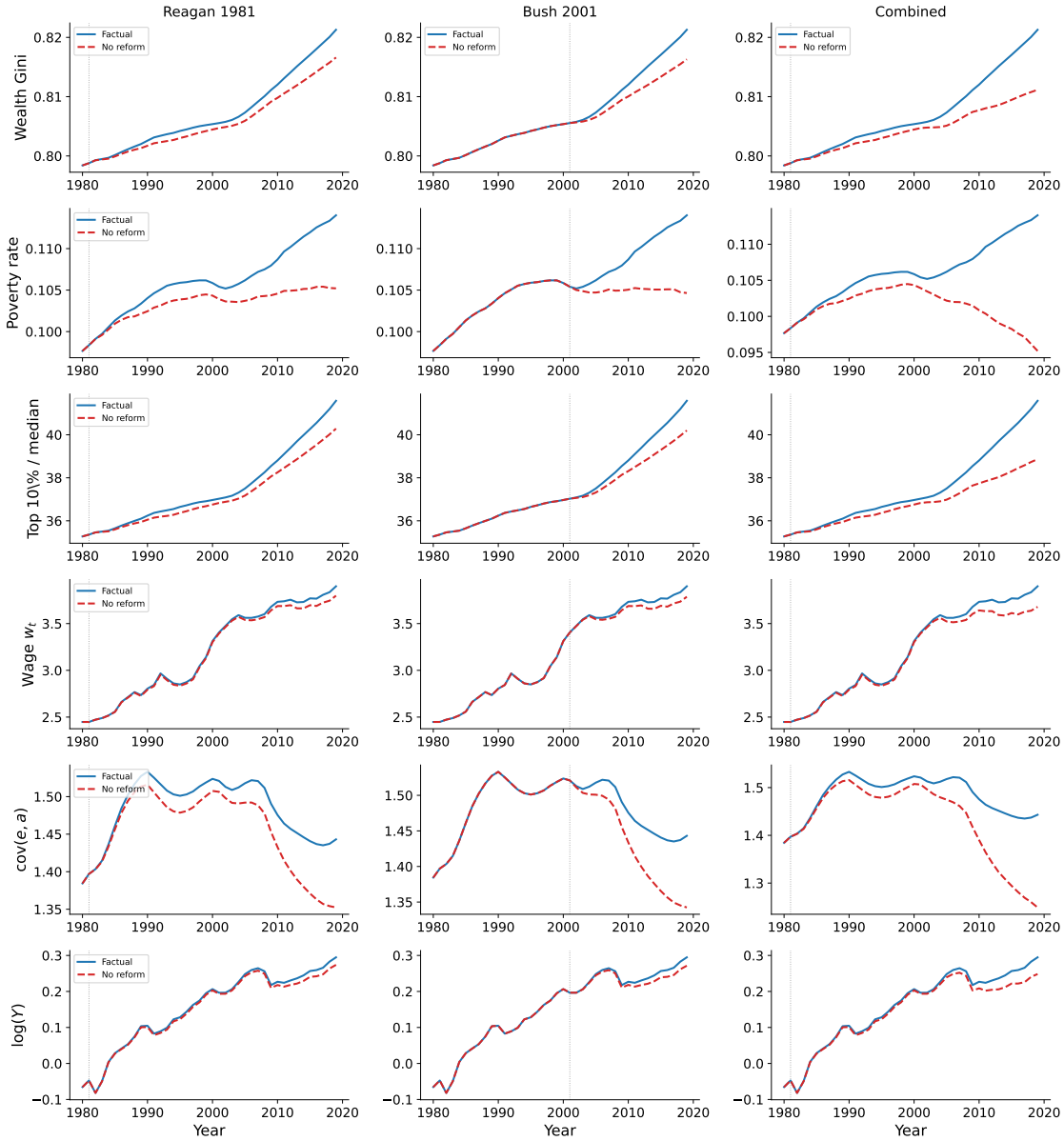

Figure 12: Macroeconomic impact of capital tax reforms

Notes: 500,000-agent simulation, 1980–2019. Columns: Reagan 1981, Bush 2001, and combined. Rows show wealth Gini, poverty rate, top 10%/median ratio, wage, return-wealth covariance, and output. Solid blue: factual; dashed red: counterfactual (no reform). Capital evolves endogenously; all GE prices re-solved each period.

that these distributional effects do not fade.

Table 9: Growth vs. inequality: summary of tax reform counterfactuals

	Reagan 1981		Bush 2001		Combined	
Δ	2010	2019	2010	2019	2010	2019
Growth (%)						
Output	+0.85	+1.75	+0.77	+1.87	+1.65	+3.75
Wage	+0.76	+1.44	+0.70	+1.55	+1.49	+3.15
Inequality measures (%)						
Wealth Gini	+0.18	+0.35	+0.15	+0.36	+0.34	+0.74
Top 10%/median	+1.12	+2.23	+0.94	+2.28	+2.08	+4.67
Poverty rate	+2.82	+4.94	+2.39	+5.11	+5.26	+10.23
Income share shift (pp)						
Top 1%	+1.62	+1.72	+1.93	+1.98	+4.06	+4.35
Top 10%	+1.85	+1.98	+2.20	+2.27	+4.62	+4.99
Bottom 50%	−0.93	−1.02	−1.10	−1.16	−2.30	−2.54
Bottom 10%	−0.15	−0.17	−0.18	−0.19	−0.37	−0.42

Notes: Growth rows: (factual − counterfactual) / counterfactual × 100. Income share shifts in percentage points. Both reforms are active by 2010. Combined stacks both reforms' offsets additively.

4.9 Scope

The application clarifies both the promise and the limits of the theory. The recovered shocks pass non-targeted validation checks, the forecasting exercise uses the belief-invariant structure in a way that reduced-form benchmarks do not, and the distributional simulation turns the analytical covariances into observable inequality dynamics. At the same time, the baseline amplification and risk-pricing magnitudes are modest. In a conservative annual calibration, the new state variables mostly explain why representative-agent aggregation works so well; they become quantitatively central when return dispersion, sorting, or the distributional state is larger.

5 Conclusion

This paper identifies the distributional objects behind approximate aggregation. In the model, aggregate capital is not the only relevant state; two sorting covariances also matter. They are sufficient for contemporaneous equilibrium allocations, they enter the aggregate capital law of motion in closed form, and they explain why representative-agent aggregation

is accurate when sorting is quantitatively small.

The result also clarifies the limits of the reduction. Future sorting is not generally closed in the three moments, so the full cross-section must still be simulated. The gain is that the simulation uses exact household policies and an exact aggregate capital recursion, rather than an estimated forecasting rule. Belief invariance further removes the fixed point over perceived aggregate dynamics within this class of preferences.

The U.S. application is best read as validation and scope. The inversion recovers sensible shocks in simulated Krusell–Smith economies, the recovered wedges line up with external labor and saving indicators, and the shock scatter reveals a two-dimensional business cycle. Belief invariance also gives a practical forecasting payoff: conditional paths can be computed without re-solving the household problem, and the rolling exercise improves on a recursive AR benchmark. At the baseline calibration, state-dependent amplification and return-risk pricing are modest. That modesty is informative: the covariances are the missing sufficient statistics, but their quantitative importance depends on the amount of return dispersion and sorting in the economy.

The broader message is that analytical aggregation need not mean representative-agent aggregation. A small number of economically interpretable distributional moments can preserve heterogeneity while making the mechanism transparent.

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